

Appendix Table A3. Full Event-Level Response Model Results

Model	Feature set	N	Classes	Accuracy	Balanced acc.	Macro F1	Top-3 acc.	JS div.
Majority	baseline	5102	10	0.203	0.100	0.034	0.523	0.6055
actor markov	actor_word_only	5102	10	0.185	0.097	0.061	0.521	0.3349
actor history markov	actor_word+previous_res	5102	10	0.448	0.382	0.412	0.641	0.0295
actor immediate markov	actor_word+immediate_o	5102	10	0.709	0.663	0.740	0.810	0.0394
Logistic actor only	actor_only	5102	10	0.093	0.117	0.091	0.315	0.1567
Hist-GBDT actor only	actor_only	5102	10	0.166	0.104	0.096	0.441	0.0731
Random Forest actor only	actor_only	5102	10	0.097	0.108	0.091	0.338	0.1108
XGBoost actor only	actor_only	5102	10	0.181	0.102	0.076	0.493	0.2150
Logistic actor history	actor_history	5102	10	0.427	0.425	0.390	0.605	0.0275
Hist-GBDT actor history	actor_history	5102	10	0.442	0.376	0.401	0.656	0.0166
Random Forest actor histo	actor_history	5102	10	0.440	0.408	0.398	0.629	0.0025
XGBoost actor history	actor_history	5102	10	0.470	0.403	0.429	0.677	0.0163
Logistic observed immedi	observed_immediate	5102	10	0.748	0.747	0.702	0.826	0.0084
Hist-GBDT observed imm	observed_immediate	5102	10	0.748	0.718	0.757	0.839	0.0058
Random Forest observed	observed_immediate	5102	10	0.759	0.751	0.731	0.823	0.0022
XGBoost observed immec	observed_immediate	5102	10	0.771	0.739	0.779	0.861	0.0068

Full actor-only, actor-history, observed-immediate, Markov, and majority model results.